Efficient learning from disjunction agreement pairs

An algorithm and proof for the formulation in Blum–Balcan (2007)

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Abstract

An elementary exclusion-closure algorithm computes the greatest disjunction consistent with labelled examples and exact agreement on unlabelled pairs. Combining this solver with two finite-class bounds gives a polynomial-time learner using $O((\ln|\mathcal{C}_{D,\chi}(\tau)| + \ln(1/\delta))/\varepsilon)$ labelled pairs and $O((n + \ln(1/\delta))/\tau)$ unlabelled pairs, for arbitrary joint distributions satisfying the realizable agreement promise. Taking $\tau = \varepsilon$ gives the labelled-sample benchmark in the stated 2007 question. A fully specified stopping rule removes the need to know or count the compatible class, with the same order of labels in expectation and with high probability.

Status. This note supplies a self-contained mathematical argument for the stated learning guarantee. It does not establish that the argument is new, that the problem remained open until this date, or that the historical reward is currently payable. Those are separate bibliographic and author-review questions. No external acceptance is claimed.

1 Setting and precise guarantee

Blum and Balcan's 2007 note [1], Section 3, asks for efficient learning with a number of labelled examples controlled by the logarithm of the nearly compatible class. The relevant item in Section 3.2 is *Co-training with disjunctions*. The reward statement in that note offers $300 for a positive solution to an item in the list.

We state the result using d fixed Boolean features $\phi_1, \dots, \phi_d$ on an object space X. The hypothesis class consists of the distinct functions

$$\mathcal{C} = \{h_S : S \subseteq [d]\}, \qquad h_S(x) = \bigvee_{i \in S} \phi_i(x).$$

The empty disjunction is 0. Repeated representations of the same Boolean function are counted only once; thus $|\mathcal{C}| \le 2^d$. Ordinary monotone disjunctions have $d = n$, with $\phi_i(x) = x_i$. Allowing negated literals is handled in Section 5.

Let D be any distribution on $X \times X$. There is a fixed $h^* = h_{S^*} \in \mathcal{C}$ such that

$$\mathbb{P}_{(x_1,x_2)\sim D}\big[h^*(x_1) = h^*(x_2)\big] = 1.$$

A labelled pair has the common label $y = h^*(x_1) = h^*(x_2)$. All sampled pairs are independent draws from D; the coordinates inside a pair can be arbitrarily dependent and can have different marginals.

Define the incompatibility probability and a prediction error by

$$q_D(h) = \mathbb{P}_D[h(x_1) \ne h(x_2)], \tag{1}$$
$$e_D(h) = \mathbb{P}_D[h(x_1) \ne h^*(x_1) \text{ or } h(x_2) \ne h^*(x_2)]. \tag{2}$$

The second quantity is stronger than error on the first view, which is the prediction convention in [1]. Either first-view or second-view error is at most $e_D(h)$. For $0 < \tau < 1$, set

$$\mathcal{C}_D(\tau) = \{h \in \mathcal{C} : q_D(h) \le \tau\}, \qquad K = |\mathcal{C}_D(\tau)|.$$

This is $\mathcal{C}_{D,\chi}(\tau)$ for the compatibility in the question. Since h^* is perfectly compatible, $1 \le K \le 2^d$.

Theorem 1 (Fixed sample sizes)**.** *Let $0 < \varepsilon, \tau < 1$ and $0 < \delta \le 1/2$. Draw u unlabelled pairs and an independent set of m labelled pairs, where*

$$u \;\ge\; \frac{d \ln 2 + \ln(2/\delta)}{\tau}, \qquad m \;\ge\; \frac{\ln K + \ln(2/\delta)}{\varepsilon}. \tag{3}$$

The exclusion-closure algorithm below returns a disjunction $\widehat{h}$ satisfying, with probability at least $1 - \delta$,

$$e_D(\widehat{h}) \le \varepsilon, \qquad q_D(\widehat{h}) \le \tau.$$

It runs in time polynomial in d, u, m. For $d \ge 1$, with dense Boolean input, a queue implementation takes $O(d(u + m) + d)$ elementary operations after the features have been evaluated.

The algorithm itself takes the samples as input and does not use K. Section 4 gives a complete rule for acquiring and stopping the labelled sample without knowing K.

2 An exact consistency solver

Write $A(x) = \{i : \phi_i(x) = 1\}$. For an unlabelled pair, write $A = A(x_1)$, $B = A(x_2)$. If $E = [d] \setminus S$ denotes excluded features, then

$$h_S(x_1) = h_S(x_2) \quad \Longleftrightarrow \quad (A \subseteq E \iff B \subseteq E). \tag{4}$$

Thus each pair gives two implications between sets:

$$A \subseteq E \implies B \subseteq E, \qquad B \subseteq E \implies A \subseteq E. \tag{5}$$

An implication fires only when *every* feature in its premise has already been excluded. An overlap with the excluded set does not suffice.

Algorithm CONS(U, L)**.**

1. Let E initially contain every feature active in either coordinate of any negatively labelled pair in L.
2. Apply all rules (5) from U, adding their conclusions to E, until no further feature can be added.
3. Set $\widehat{S} = [d] \setminus E$ and return $h_{\widehat{S}}$. For a general consistency-decision interface, check all positive labelled pairs and report inconsistency if one is not satisfied. Under the realizability promise this check always succeeds.

Lemma 2 (Safety of exclusion)**.** *Throughout the algorithm, $E \cap S^* = \varnothing$. Consequently $S^* \subseteq \widehat{S}$, and the returned disjunction has no false negatives anywhere on X.*

Proof. If a pair has label 0, neither of its coordinates can contain an active feature from S^*. Initialization therefore excludes no target feature.

Suppose the invariant holds before a rule fires, and suppose $A \subseteq E$. Then $A \cap S^* = \varnothing$, so $h^*(x_1) = 0$. Compatibility of the target on this sampled pair implies $h^*(x_2) = 0$, hence $B \cap S^* = \varnothing$. Adding B to E preserves the invariant. The argument for the other direction is identical.

All sampled pairs satisfy the target promise almost surely, including in the finite capped procedure used later. Finally $S^* \subseteq \widehat{S}$ implies $h^*(x) \le h_{\widehat{S}}(x)$ for every $x \in X$. □

Lemma 3 (Consistency and maximality)**.** *The algorithm terminates, agrees on every pair in U, and satisfies every labelled pair in L. Its support is the greatest feasible support, not just a maximal one. On arbitrary labelled input, failure of its positive-label check proves that no feasible disjunction exists.*

Proof. Each nontrivial update adds at least one feature to E, so at most d such updates occur. At termination both implications in (5) hold for every pair, proving empirical agreement.

Every feature in a negatively labelled pair was excluded initially, so both coordinates receive label 0. If a pair has label 1, each coordinate has an active target feature. By Lemma 2 these features are retained, so both coordinates receive label 1.

For maximality, replace S^* in the safety proof by any support T that agrees on U and avoids every negative labelled example. The same induction shows $T \subseteq \widehat{S}$. The returned support itself has those properties. It is therefore their greatest feasible support. Positive constraints are monotone in the support: if any feasible support satisfies a positive example, so does $\widehat{S}$. If $\widehat{S}$ fails a positive constraint, every smaller feasible support fails it as well. □

2.1 Why the computation is efficient

A simple implementation scans all pairs after each change and already takes polynomial time, at most $O(d^2u + dm + d)$ on dense input. A queue implementation is linear in the total input incidence size:

1. For each of the $2u$ pair sides, store the number of its features not yet excluded.
2. For each feature, store the pair sides in which it occurs.
3. When a feature is excluded for the first time, decrement all its incidence counters. Enqueue a side when its counter reaches zero.
4. On removing a zero-count side from the queue, exclude all features on the opposite side of that pair.

Each feature is excluded at most once; each side reaches zero at most once; and each occurrence is used in at most one decrement and one opposite-side scan. All sides whose counters are initially zero, including empty sides, are enqueued. Preprocessing and checking labelled input are also linear in its incidence size.

Equivalently, putting $z_i = 1$ when feature i is excluded, a pair contributes Horn implications

$$\bigwedge_{i \in A} z_i \implies z_j \quad (j \in B), \qquad \bigwedge_{j \in B} z_j \implies z_i \quad (i \in A).$$

The queue groups the common premises, avoiding an unnecessary expansion into $|A||B|$ repeated literal occurrences. No satisfiability or counting oracle is required.

3 Proof of the labelled-sample guarantee

For the unlabelled sample U, define the empirical compatible class

$$\mathcal{H}_U = \{h \in \mathcal{C} : h(x_1) = h(x_2) \text{ for every } (x_1, x_2) \in U\}.$$

This class is only used in the proof; the learner never enumerates it.

Lemma 4 (Unlabelled reduction)**.** *For every $u \geq 0$,*

$$\mathbb{P}[\mathcal{H}_U \not\subseteq \mathcal{C}_D(\tau)] \leq |\mathcal{C}|e^{-\tau u}.$$

Proof. For a fixed h with $q_D(h) > \tau$, the probability of agreeing on all u independent pairs is

$$(1 - q_D(h))^u \leq e^{-q_D(h)u} \leq e^{-\tau u}.$$

Sum this bound over the at most $|\mathcal{C}|$ hypotheses outside $\mathcal{C}_D(\tau)$. □

Lemma 5 (Conditional labelled bound)**.** *For a fixed U, if L consists of m independent labelled pairs drawn independently of U, then*

$$\mathbb{P}[\exists h \in \mathcal{H}_U : e_D(h) > \varepsilon \text{ and } h \text{ satisfies } L \mid U] \leq |\mathcal{H}_U|e^{-\varepsilon m}.$$

Proof. The class $\mathcal{H}_U$ is fixed after conditioning on U. A fixed hypothesis of pair prediction error p satisfies an independent labelled pair on both coordinates with probability $1 - p$. Its probability of satisfying all m pairs is $(1 - p)^m \leq e^{-pm}$. Sum over $\mathcal{H}_U$. No independence of the two coordinates is used. □

Proof of Theorem 1. By the first inequality in (3) and Lemma 4, with probability at least $1 - \delta/2$ the event

$$\mathcal{E}_U = \{\mathcal{H}_U \subseteq \mathcal{C}_D(\tau)\}$$

holds. On that event $|\mathcal{H}_U| \leq K$, and every possible algorithmic output has incompatibility at most τ.

The solver returns a member of $\mathcal{H}_U$ satisfying all labels, by Lemma 3. Conditional on any such U, Lemma 5 and the second inequality in (3) give

$$\mathbb{P}[e_D(\widehat{h}) > \varepsilon \mid U] \leq Ke^{-\varepsilon m} \leq \delta/2.$$

Combining the two failure probabilities proves both asserted error bounds. The algorithmic analysis proves the running-time claim. □

In particular, for $\tau = \varepsilon$ and $d = n$ or $2n$,

$$\boxed{u = O\left(\frac{n + \ln(1/\delta)}{\varepsilon}\right), \qquad m = O\left(\frac{\ln|\mathcal{C}_{D,\chi}(\varepsilon)| + \ln(1/\delta)}{\varepsilon}\right).}$$

The use of unlabelled data is substantive: the logarithm in the labelled bound is that of the nearly compatible class, not the full 2^d class.

4 Stopping without knowing the compatible class

The fixed-sample theorem does not require counting K to run the solver. To remove even the need to choose a label budget using an unknown K, the following fully specified wrapper uses fresh random labels and independent validation. It does not make active label queries or select examples for labelling.

Let

$$N = \left\lceil \frac{d \ln 2 + \ln(4/\delta)}{\tau} \right\rceil, \quad b = \ln(16/\delta), \quad m_j = \left\lceil \frac{64\, 2^j b}{\varepsilon} \right\rceil \quad (j \geq 0). \tag{6}$$

Draw N unlabelled pairs once. The computable final phase is

$$J = \min \left\{ j \geq 0 : m_j \geq \frac{4}{\varepsilon}(d \ln 2 + b) \right\}.$$

The constant 64 is deliberately conservative; J is allowed to be 0.

For $j = 0, \ldots, J$:

1. Draw m_j fresh labelled training pairs and compute $h_j = \mathrm{CONS}(U, L_j)$.
2. If $j = J$, return h_j.
3. Otherwise draw m_j fresh labelled validation pairs. Return h_j if the fraction on which at least one view is misclassified is at most $\varepsilon/2$. If not, continue.

Theorem 6 (Automatic label budget)**.** *The wrapper always terminates in polynomial time. With probability at least $1 - \delta$, it simultaneously returns a hypothesis with*

$$e_D(\widehat{h}) \leq \varepsilon, \qquad q_D(\widehat{h}) \leq \tau$$

and uses at most

$$Q \leq 400\, \frac{\ln K + \ln(16/\delta)}{\varepsilon} \tag{7}$$

labelled pairs, counting both training and validation. Moreover,

$$\mathbb{E}Q = O\left(\frac{\ln K + \ln(1/\delta)}{\varepsilon} \right).$$

A deterministic worst-case bound uses $d \ln 2$ instead of $\ln K$. The bound through K is a high-probability and an expectation guarantee.

4.1 Accuracy and high-probability label count

The event $\mathcal{E}_U$ now fails with probability at most $\delta/4$. Conditional on a current hypothesis and all prior information, its validation block remains independent. Standard binomial bounds give:

$$\begin{aligned} e_D(h_j) > \varepsilon \quad &\Longrightarrow \mathbb{P}[\text{validation accepts}] \leq e^{-\varepsilon m_j/8}, \\ e_D(h_j) \leq \varepsilon/4 \quad &\Longrightarrow \mathbb{P}[\text{validation rejects}] \leq e^{-\varepsilon m_j/12}. \end{aligned}$$

The first is the lower-tail bound at half the mean. For the second, binomial stochastic domination reduces to mean $\varepsilon/4$, and the upper-tail bound is applied at twice that mean.

The probability of ever accepting an error greater than ε before the final phase is at most

$$\sum_{j \geq 0} e^{-\varepsilon m_j/8} \leq \sum_{j \geq 0} e^{-8\, 2^j b} \leq 2e^{-8b} \leq \delta/16.$$

At phase J, Lemma 5 over the entire class gives

$$\mathbb{P}[e_D(h_J) > \varepsilon/4 \mid U, \text{past}] \le 2^d e^{-\varepsilon m_J/4} \le e^{-b} = \delta/16.$$

These bounds do not require $\mathcal{E}_U$.

For analysis only, define the unknown phase

$$j^* = \min\left\{ j \ge 0 : m_j \ge \frac{4}{\varepsilon}(\ln K + b) \right\}.$$

Since $K \le 2^d$, $j^* \le J$. Conditional on $\mathcal{E}_U$, if phase j^* is reached, its fresh training sample yields $e_D(h_{j^*}) \le \varepsilon/4$ except with probability $e^{-b} = \delta/16$. If $j^* < J$, rejection of such a hypothesis has probability at most $e^{-\varepsilon m_{j^*}/12} \le e^{-b} = \delta/16$. If $j^* = J$, it stops regardless.

Therefore the probability, on $\mathcal{E}_U$, of continuing beyond j^* is at most $\delta/8$. Conditioning on histories is legitimate because each training and validation block is fresh; alternatively, all independent blocks can be generated in advance.

On stopping by j^*, the number of labels obeys

$$Q \le 2\sum_{j=0}^{j^*} m_j \le 6m_{j^*} \le 384\,\frac{\ln K + b}{\varepsilon} + 6 \le 400\,\frac{\ln K + b}{\varepsilon}.$$

For the middle inequalities, use $\lceil c2^j \rceil \le c2^j + 1$, $j^* + 1 \le 2^{j^*}$, and minimality of j^*. If $j^* = 0$, $m_0 \le 64b/\varepsilon + 1$; otherwise $m_{j^*} < 8(\ln K + b)/\varepsilon + 1$.

The total failure probability for the two errors and the label bound is at most

$$\delta/4 + \delta/16 + \delta/16 + \delta/8 = \delta/2 \le \delta.$$

Replacing K by 2^d in the same size calculation bounds the deterministic cap. In particular, $J = O(\log(1 + d/\ln(1/\delta)))$, and all time and sample bounds are polynomial in the ordinary learning parameters.

4.2 Expected label count

For a fixed U, put $k_U = |\mathcal{H}_U| \ge 1$ and define j_U by replacing K with k_U in the definition of j^*. At every reached phase $j \ge j_U$, the conditional probability of continuing is at most

$$r = 2e^{-b} = \delta/8 \le 1/8.$$

This follows from the same training and validation bounds; at the final phase continuation probability is zero. Also $m_{j+1} \le 2m_j$. Consequently the expected cost after phase j_U is at most

$$\sum_{s \ge 1} 2^{s+1} m_{j_U} r^s = 2m_{j_U}\frac{2r}{1-2r} \le \frac{2}{3}m_{j_U}.$$

Together with the cost of earlier phases,

$$\mathbb{E}[Q \mid U] \le 7m_{j_U} \le 448\,\frac{\ln k_U + b}{\varepsilon} + 7.$$

It remains to control rare, unhelpful unlabelled samples without paying an extra term proportional to δd. Every $h \notin \mathcal{C}_D(\tau)$ belongs to $\mathcal{H}_U$ with probability at most $e^{-\tau N}$. Hence

$$\mathbb{E}k_U \le K + |\mathcal{C}|e^{-\tau N} \le K + \delta/4.$$

Jensen's inequality gives

$$\mathbb{E}\ln k_U \le \ln(K + \delta/4) \le \ln K + \delta/4.$$

Substitution proves the stated expectation bound. This also shows why no class-counting procedure is hidden inside the stopping rule.

5 Literal conventions and extensions

5.1 Negated literals

For ordinary Boolean disjunctions with negations, use the feature map

$$\phi(x) = (x_1, \dots, x_n, 1 - x_1, \dots, 1 - x_n), \qquad d = 2n.$$

The consistency proof applies to this restricted set of feature vectors without change. It never assumes that all 2^{2n} feature vectors can occur.

The counting in K is over distinct Boolean functions. In particular, every literal subset containing both x_i and $\neg x_i$ represents the same constant-one function. One can canonicalize all such outputs to 1. The full signed disjunction class, including the empty disjunction, has at most $3^n + 1$ distinct functions; the looser bound 2^{2n} is sufficient for the unlabelled sample.

If a convention excludes constant functions, adjoining them changes K by at most two, so the order of the labelled bound is unchanged. In that optional convention the output may be outside the restricted class. The principal theorem uses the class of all disjunctions generated by the stated literals, including the empty disjunction and tautologies when representable.

5.2 Different disjunctions in two views

The same argument also covers targets h_1^*, h_2^* on two different feature sets, promised to agree on the pair distribution. Use disjoint feature indices for the two views, take the two active sets in that common index universe, and apply the same two implications. The compatible-class count then refers to distinct hypothesis pairs. No conditional-independence assumption is needed for this extension.

6 Small example and verification

Consider five features and the unlabelled pairs

$$(\{2\}, \{3, 4\}), \qquad (\{4\}, \{5\}), \qquad (\{1, 3\}, \{1, 5\}).$$

Suppose a labelled negative pair is $(\{2\}, \{2\})$. Initially $E = \{2\}$. The first agreement pair forces $3, 4$ into E; the second forces 5 into E. The third pair imposes no further exclusion. The output is $\widehat{h}(x) = x_1$. A positive labelled pair $(\{1, 3\}, \{1, 5\})$ is satisfied automatically.

The accompanying code implements the queue solver, signed feature lifting, and the fully specified wrapper. An independently written finite enumerator checked the underlying closure property on 19,560 systems: 5,560 exhaustive two-pair systems with all exclusion masks for $d \leq 3$; 10,000 random systems for $4 \leq d \leq 8$; and 4,000 target-generated labelled systems. It checked 48,000 labelled pairs in the last group. No counterexample was found.

A separate verifier compared the actual queue implementation against complete enumeration on 22,560 systems, including 2,000 signed-literal systems and 1,000 arbitrary, potentially inconsistent systems. All 160,627 feasible-support witnesses and all 71,335 deletion-certificate steps passed. It correctly rejected 746 inconsistent systems. Exact results are in `solver_verification_results.json`.

These are finite consistency checks, not a machine-checked formalization of the probabilistic theorems. The mathematical proofs above establish the all-dimensions guarantee.

7 Bibliographic and reward status

The 2013 paper [3] studies two-sided disjunctions, where two OR functions give complementary labels to a *single* object, with additional distributional conditions in its passive results. The 2024 paper [4] studies co-training under a weak-dependence condition using active learning. Neither set of hypotheses and resource bounds directly supplies the arbitrary-distribution, passive guarantee proved here.

The later author version of the framework paper [2], Section 6.2, does not repeat the particular co-training-disjunction item, though it does list other related questions. This omission does not establish that the item had been solved, but it prevents treating the old list alone as proof of continuing openness.

The present search did not establish a previous publication of the specific exclusion-closure argument, a payment of this particular reward, or a current renewal of the reward. Establishing priority and applying the 2007 monetary promise require review by the problem's authors. The mathematical theorem here and those administrative questions should be evaluated separately.

Preparation and reproducibility. This note was prepared with GPT-6 Astra assistance for Grisha Pochuev. Independent reasoning passes checked the finite solver and the adaptive sample argument. The supplied Python code requires only the standard library. Run `python cotrain_disjunctions.py -demo` for the example and `python verify_solver.py` for the implementation checks. The source LaTeX and exact verification results accompany the PDF.

References

[1] A. Blum and M.-F. Balcan. *Open Problems in Efficient Semi-Supervised PAC Learning*. COLT 2007, pp. 622–624, especially Sections 3 and 3.2. `https://home.ttic.edu/~avrim/Papers/openproblems.pdf`

[2] M.-F. Balcan and A. Blum. *A Discriminative Model for Semi-Supervised Learning*. Journal of the ACM 57(3), 2010. Author version (2009). `https://home.ttic.edu/~avrim/Papers/labeled_unlabeled-2009.pdf`

[3] M.-F. Balcan, C. Berlind, S. Ehrlich, and Y. Liang. *Efficient Semi-supervised and Active Learning of Disjunctions*. ICML 2013, PMLR 28(1), pp. 633–641. `https://proceedings.mlr.press/v28/balcan13.html`

[4] I. Diakonikolas, M. Ma, L. Ren, and C. Tzamos. *Fast Co-Training under Weak Dependence via Stream-Based Active Learning*. ICML 2024, PMLR 235, pp. 10841–10864. `https://proceedings.mlr.press/v235/diakonikolas24b.html`